

SHRESHT ASHISH

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About

Computer Science and Engineering student with a focus on information retrieval and retrieval-augmented generation, backed by a broader foundation in machine learning and full-stack development. Experienced in building hybrid retrieval and RAG pipelines.

Education

Institute of Technology, Nirma University

Bachelors of Technology, Computer Science and Engineering

Ahmedabad, Gujarat, India

2025–2029

Experience

Amnex Infotechnologies

Intern - Mobility

Ahmedabad, Gujarat, India

June 2026

- Implemented a two-stage retrieve-then-rerank pipeline with a **cross-encoder reranker** and title-aware semantic table discovery, plus **LLM-based reconstruction** of malformed PDF tables.
- Built the agent-orchestration loop for AIRA an HR assistant: **Gemma 3 27B** reasons over a **constrained JSON action schema** (`call_tool` / `answer` / `clarify` / `escalate`) across up to **4 tool round-trips**, with a **confidence-gated synthesis** step that escalates to HR below a **0.6 threshold** and falls back to extractive passage retrieval if the LLM is unavailable.
- Built a 6-API sync client for an external HR platform (per-API field mappers, hierarchical **RBAC**) and the full frontend in **Next.js 16/Tailwind CSS v4**, including email-OTP authentication and role-aware dashboards.
- Reverse-engineered** a legacy government pollution-data portal's export feature, discovering a predictable URL pattern for on-demand **.xlsx exports** across arbitrary regions and timeframes – enabling full programmatic scraping in place of manual UI-driven collection; containerized the scraper (**Docker**) with automated **CAPTCHA handling** and **asynchronous background ingestion**.

Projects

ML/RAG Race Analytics Platform (XGBoost, Qdrant, FastAPI, Next.js)

2026

- Built a two-stage pipeline: an **XGBoost regressor** predicts pace deltas from telemetry, feeding as an engineered **feature** into a downstream **XGBoost classifier** predicting podium probability.
- Evaluated using a **temporal train/test split** (2022–2023 train, 2024 test) rather than random splitting, to prevent look-ahead leakage and mirror deployment on an unseen season.
- Built a **RAG** layer for natural-language Q&A over race data, combining dense vector search with structured metadata filters via **Qdrant** for hybrid retrieval, using **asymmetric embeddings** for documents vs. queries.
- Delivered full-stack via **FastAPI** and **Next.js/TypeScript**, with a unidirectional flow from query to RAG answer to rendered dashboard.

ArogyaLink (Django, PostgreSQL, Gemma 4, AI4Bharat)

Unleash LLM Hackathon | 2026

- Owned the end-to-end **AI module** of a Django-based rural healthcare platform, designing a pipeline that converts **multilingual voice narration** and **photographed paper health registers** into **structured clinical records** for India's **Swasth Bharat Portal** (ASHA/ANM/CHO worker workflows).
- Built the speech-to-text stage on AI4Bharat's **indic-conformer-600m-multilingual** model with an **RNNT decoder**, deployed as a singleton for efficient reuse across requests.
- Integrated **Gemma 4** (via Google AI Studio) for **multimodal structured extraction** for mapping ASR transcripts and OCR'd register photos directly onto government-mandated clinical fields, with explicit human-in-the-loop verification before submission.
- Architected a **deterministic rule engine** to isolate clinical decision logic (CBAC risk scoring, PHQ-2 depression screening) from the LLM entirely.

Skills

Programming Languages: Python, JavaScript, TypeScript, HTML, CSS, C, C++

Web Development: MERN Stack (MongoDB, Express.js, React.js, Node.js), Next.js, Tailwind CSS, Zustand, jQuery

Machine Learning & AI: TensorFlow, Keras, Scikit-learn, Pandas, XGBoost, Retrieval-Augmented Generation (RAG), LangGraph

Backend & Infrastructure: FastAPI, PostgreSQL, pgvector, Qdrant, Docker, RESTful APIs

Concepts: Artificial Intelligence, Information Retrieval, Natural Language Processing, Cybersecurity, Data Clustering, Distributed Systems

Certifications & Courses

Machine Learning Specialization

Certificate

The Web Developer Bootcamp 2026

Certificate

Google Cybersecurity Professional Certificate

Certificate